
Behavioral Conditioning of Video World-Model Predictors for Egocentric Action Anticipation

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Code: https://github.com/ORG/REPO • Project: EgoProject 2026 (Univ. Rostock × UBB Cluj)

Abstract

[Abstract placeholder — write last.]

1 Introduction

[lead: Parsa + Erfan (merge)] [Draft pending.]

2 Background and related work

[lead: Erfan (JEPA/VL-JEPA); Parsa (gaze-conditioning refs)] [Draft pending.]

3 Method

[lead: Parsa]

3.1 Setup: frozen-representation probing

[Draft pending.]

3.2 Conditioning pathways: weak vs. strong

[Draft pending.]

3.3 Direct (PEVA-style) readout

[Draft pending.]

4 Experiments

[lead: Parsa (controls/results); Erfan (horizon table)]

Table 1: Placeholder main comparison. Replace with locked numbers; add a one-line take-away to the caption.

Condition	Tokens	Seed 42	Seed 43	Seed 44
6a encoder only	1568	–	–	–
6b + zeros	1764	–	–	–
6c + last block repeat	1764	–	–	–
5 + behavioral pred.	1764	–	–	–

4.1 Data and protocol

[Draft pending.]

4.2 Controls and confounds

[Draft pending.]

4.3 Results

[Draft pending.]

5 Discussion

[lead: Parsa] [Draft pending.]

6 Limitations

[lead: Parsa] [Draft pending.]

7 Conclusion

[lead: Parsa + Erfan] [Draft pending.]

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A Contribution statement

The authors contributed to distinct, complementary workstreams under a shared research goal. Individual contributions are detailed below.

Seyyed Parsa Sharifi. [Fill in.]

Erfan Yekehzare. [Fill in.]

Ioana Marica. [Fill in.]

B Extended experimental details

B.1 Full Phase 5/6 seed table

[Fill from EXPERIMENT_LOG_COMPLETE.md.]

B.2 PEVA stage-1 and stage-2 full statistics

[Fill.]

B.3 Verb/noun dissociation — all horizons, v2 vs v3, with CIs

[Fill.]

B.4 $d \gg n$ comparator diagnostic

[Fill.]

C Selected code

C.1 WEAK-equivalence check (bit-identical to Phase 5)

[Insert snippet.]

C.2 C5 gradient harness (positive check)

[Insert snippet.]

C.3 STRONG key-bias hook (DESIGNED — NotImplementedError)

[Insert snippet.]

C.4 Batch-dict schema

[Insert snippet or small table.]

D Abbreviations and notation

Term	Meaning
JEPA	Joint-Embedding Predictive Architecture
V-JEPA 2	Video JEPA 2 (frozen encoder + latent predictor)
V-JEPA 2-AC	... Action-Conditioned variant (robot signals)
V-JEPA 2.1	Dense-feature V-JEPA revision
ViT-L / ViT-G	Vision Transformer, Large / Giant
EK100	EPIC-KITCHENS-100
HD-EPIC	Highly-Detailed EPIC egocentric dataset
R@5	Recall at 5
RoPE	Rotary Position Embedding
PEVA	[VERIFY expansion] — direct real-vs-masked readout
WEAK	Peer-token concatenation pathway (built)
STRONG	GazeQwen-informed fusion pathway (designed, not built)
6a/6b/6c	Control family (encoder-only / +zeros / +repeat)
CI	Confidence interval